Lemma. Let $M \subset \mathbb{R}^3$ be a surface, and let $\alpha : I \to M$ be a curve.
If for some patch $X : D \to \mathbb{R}^3$, range $\alpha \subseteq X[D]$,
then there exist unique differentiable functions a_1, a_2 on I such that

$$\alpha(t) = X(a_1(t), a_2(t)) \text{ for all } t \in I.$$

Proof. By definition